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An Ensemble Machine Learning and Explainable AI Framework for Thyroid Disease Classification

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***Abstract*— Thyroid disorders are one of the most common endocrine diseases in the world. Early identification is crucial for effective treatment. However, diagnosing patients can be difficult as symptoms often overlap with other conditions. To address these issues, this study proposes an ensemble machine learning framework for reliable classification of thyroid conditions into negative and sick categories. The approach integrates random oversampling (RO) and random undersampling (RUS) with diverse feature optimization strategies, including feature selection methods like mRMR, RFE, feature importance methods such as DT, XGB, and feature reduction techniques such as PCA, LDA. Multiple experiments were conducted using logistic regression, random forest, support vector machine, k-nearest neighbors, and extreme gradient boosting within a tuned ensemble model. Ten-fold cross-validation ensured a detailed evaluation. Explainable AI techniques were applied to highlight the influence of key features on the result. Experimental results demonstrate that the proposed method provides excellent classification. It has an accuracy, precision, recall, and F1 score of 0.999. The best results were obtained using soft voting, outperforming existing approaches in thyroid disease prediction.**

Keywords—Thyroid disease, Ensemble model, Feature optimization, Oversampling, Undersampling, Explainable AI.

I. INTRODUCTION

Thyroid disorders are among the most common endocrine conditions worldwide, affecting millions of people each year. Global health studies indicate that hypothyroidism affects about 4–8% of the population in iodine-sufficient regions, while hyperthyroidism is found in 1–3% of individuals, particularly among women and older adults [1]. If left untreated, thyroid dysfunction may lead to severe complications such as cardiovascular disease, infertility, and reduced quality of life. Despite its prevalence, thyroid disease remains difficult to diagnose at an early stage because its symptoms overlap with those of other illnesses and many diagnostic features lack clear discriminative power. These challenges have encouraged the development of computer-aided diagnostic systems that can assist physicians in the accurate prediction of thyroid disease [2].

In recent years, machine learning (ML) has played an important role in medical fields, including disease prediction [3]. For thyroid disease, models such as support vector machines (SVM), random forests (RF), logistic regression (LR), k-nearest neighbors (KNN), and extreme gradient boosting (XGB) techniques have delivered good results [4]. However, imbalanced data distribution between healthy and sick classes often skews models toward the majority class. Additionally, classifiers are prevented from obtaining high predictive accuracy due to unnecessary features [5].

To address these issues, researchers have explored different sampling techniques and feature optimization methods. The number of samples in the minority class can be increased using Random Oversampling (RO). On the other hand, the number of samples in the majority class can be decreased using Random Undersampling (RUS). Feature optimization methods are used to enhance the performance of the classification by selecting important features. Techniques like minimum redundancy maximum relevance (mRMR) and recursive feature elimination (RFE) select the most useful features. Methods like decision tree (DT) and extreme gradient boosting (XGB) are used for determining feature importance. Dimensionality reduction methods including principal component analysis (PCA) and linear discriminant analysis (LDA) transform the feature space into more compact and discriminative representations [6]. Although existing studies have applied these strategies, most have evaluated them individually, and systematic comparisons that integrate both sampling and feature optimization within an ensemble learning framework remain limited.

The major contributions of this paper are summarized as follows:

- An ensemble-based machine learning framework is proposed, combining sampling and feature optimization for thyroid disease prediction.

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- Fifteen experimental cases are designed by integrating RO and RUS with feature selection, feature importance, and feature reduction methods.
- The ensemble incorporates LR, RF, SVM, KNN, and XGB with hyperparameter tuning, validated using 10-fold cross-validation.
- Explainable artificial intelligence (XAI) is applied to analyze feature contributions and enhance interpretability.
- Both soft and hard voting strategies are investigated, with soft voting achieving superior results.

II. Literature Review

This section provides an overview of various research studies that have utilized machine learning (ML) methods to detect and classify thyroid disorders, employing clinical datasets.

Uddin et al. [7] focused on predicting thyroid disease using an ensemble learning pipeline that combined hybrid feature selection and class-imbalance handling. The approach evaluated five machine-learning models alongside a hard voting ensemble built over Random Forest (RF) and Decision Tree (DT), after selecting features with XGBoost and SelectKBest, and mitigating imbalance via random oversampling. The results indicated that the ensemble obtained the strongest performance, yielding 100% sensitivity with an overall accuracy of 99.71%, while the individual learners trailed the ensemble. Raza et al. [8] focused on improving interpretable thyroid disease diagnosis by coupling synthetic oversampling with tree-based learners. The study was done on a dataset of 3,772 observations and applied SMOTE-NC to address categorical numeric model behavior explained using SHAP analysis. The results showed that the SMOTE-NC + LightGBM configuration obtained strong performance, with an accuracy of 0.96. Zhang et al. [9] proposed a model based on CNN for classifying thyroid disorders. An accuracy of 0.972 was obtained for the ultrasound and an accuracy of 0.942 for the CT scan image dataset using Xception. Chaganti et al. [10] used SVM, KNN, DT, NB, and RF on GitHub datasets, which obtained 90% accuracy for classifying thyroid disorders. Kumar et al. [11] proposed a DeepCNN with Gray Wolf Optimizer (GWO) model, which achieved 95% accuracy and 92% specificity for classifying thyroid disorders. Pal et al. [12] proposed a framework using KNN, DT, and MLP on the UCI dataset for classifying thyroid disorders. MLP achieved 95.73% accuracy and 94.23% AUC. Alyas et al. [13] used DT, RF, KNN, and ANN on the UCI thyroid dataset using both sampled and unsampled data and showed the comparison. RF obtained the highest accuracy of 94.8% and specificity of 91%.

III. Methodology

This study followed a structured methodology consisting of dataset preprocessing, class imbalance treatment, feature optimization, and classification through ensemble learning, as shown in Fig. 1. The dataset was collected from Kaggle [14], containing 3772 records with 30 attributes, of which 3541 were negative and 231 were sick cases. The methodology was designed to ensure balanced learning, optimized feature representation, and robust evaluation through 10-fold cross-validation.

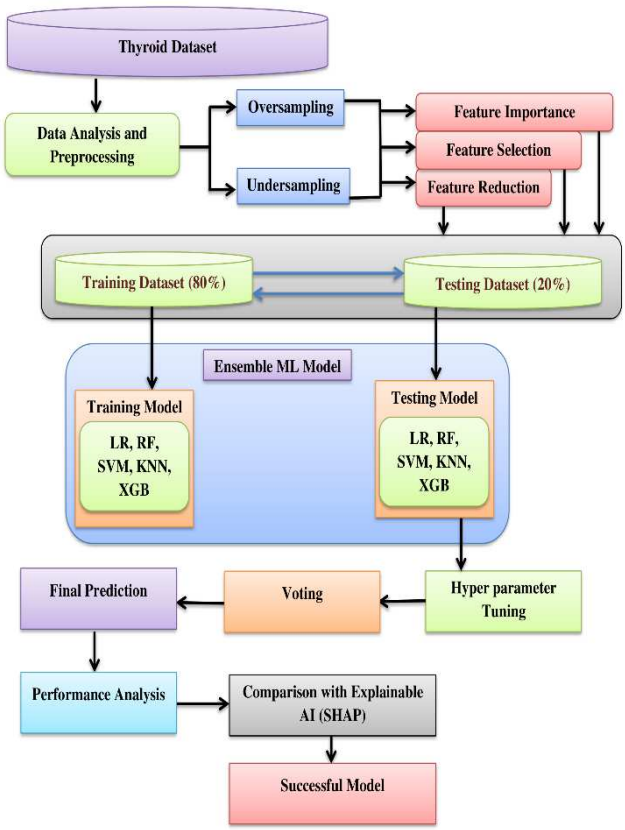


Fig. 1. Architecture of the proposed model

A. Dataset Description

The dataset contains patient records that are labeled as either sick or negative. It has continuous and categorical features. These features include sex, medication status, pregnancy, thyroid surgery, I131 treatment, diagnostic queries, goitre, tumor, hypopituitarism, psychiatric history, TSH, T3, TT4, T4U, FTI, TBG, and referral source codes.

B. Dataset Preprocessing

Preprocessing was performed to ensure data quality and consistency. Text attributes were converted into numerical values, such as true/false encoded as 1 and 0. Referral sources were encoded as: WEST = 0, STMW = 1, SVHC = 2, SVI = 3, SVHD = 4, and other = 5. The decision column was mapped to 0 for negative and 1 for sick cases. Missing values, originally represented as "?", were imputed using median values for robustness. Normalization and cleaning were also applied, producing a final dataset of 28 attributes suitable for experimentation.

C. Class Imbalance Treatment

The dataset exhibited significant class imbalance, with sick cases representing only 6.1% of the total. To address this, Random Oversampling (RO) and Random

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Undersampling (RUS) were applied in addition to the original dataset as baseline. RO increased the frequency of minority class samples, while RUS balanced the dataset by reducing majority class samples. These methods enabled the exploration of data-driven effects on classifier performance.

D. Feature Optimization

Three categories of feature optimization were employed: feature selection, feature importance, and feature reduction. Each category incorporated two techniques, resulting in six optimization schemes for both RO and RUS.

Feature Selection: minimum redundancy maximum relevance (mRMR) and Recursive Feature Elimination (RFE) were applied to select relevant subsets of features.

Feature Importance: Decision Tree (DT) and Extreme Gradient Boosting (XGB) were used to rank features by contribution.

Feature Reduction: Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) were used to reduce dimensionality while retaining important information.

This design, combined with baseline cases (original, RO only, and RUS only), produced a total of 15 experimental cases.

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E. Classification

Classification was carried out using an ensemble learning framework constructed from five base classifiers: Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbor (KNN), and XGB. Hyperparameter tuning was applied to all models to ensure optimized performance. For hyperparameter optimization, we employed RandomizedSearchCV with 3-fold cross-validation. The parameter ranges for each model are listed in Table I. This approach was chosen over exhaustive grid search to reduce computational overhead while maintaining good coverage of the hyperparameter space. The best parameters selected for each classifier are reported in Section IV. The tuned models were aggregated using both soft voting and hard voting ensembles. However, ensemble voting strategies were selectively tested on the top-performing cases to evaluate improvements. Model validation was carried out using 10-fold cross-validation, and Explainable Artificial Intelligence (XAI) was applied to the best-performing ensemble to provide feature attribution insights.

IV. Results

A. Evaluation Matrix

The confusion matrix is a tabular representation of the classification model's performance, showing the counts of correctly and incorrectly classified instances for each class in a binary classification setting with negative and sick categories. It enables the computation of key performance indicators and helps identify areas for model refinement. In this study, four standard metrics were used: accuracy, precision, recall, and the F1-score. These metrics provide a comprehensive evaluation of model effectiveness and are widely adopted in clinical decision-support research.

Let TP denote the number of true positives, TN the number of true negatives, FP the number of false positives, and FN the number of false negatives. The metrics are defined as follows:

Accuracy represents the overall correctness of the model:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (1)$$

Precision measures the proportion of predicted positive cases that are actually positive:

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

Recall indicates the proportion of actual positive cases correctly identified:

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

The F1-score provides a balanced measure of precision and recall:

$$F1\ Score = \frac{2 \times (Precision \times Recall)}{Precision + Recall} \quad (4)$$

All metrics were computed for each of the fifteen experimental cases using ten-fold cross-validation to ensure robust and unbiased evaluation. The confusion matrix of the best-performing experimental case is shown in Fig. 2.

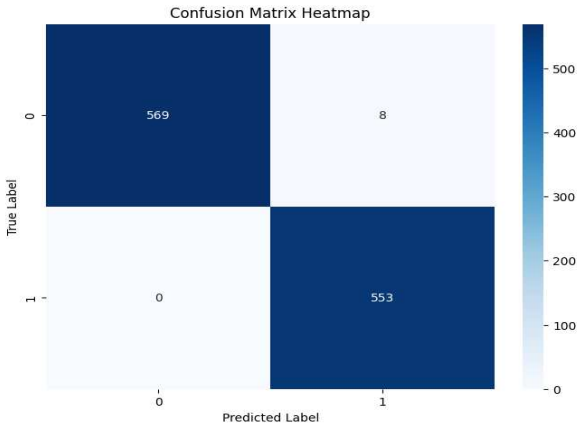


Fig. 2. Confusion Matrix of the best performed experimental case.

B. Comparative Analysis of Sampling and Feature Optimization

The performance of five baseline single classifiers was first evaluated on the preprocessed dataset, as presented in Table I. Logistic Regression (LR) achieved 0.960 accuracy, with precision and recall of 0.96 and 0.95. Random Forest (RF) and Extreme Gradient Boosting (XGB) performed best, reaching 0.980 across all metrics. Support Vector Machine (SVM) and k-Nearest Neighbors (KNN) showed weaker results, with accuracies of 0.936 and 0.934, and F1-scores of 0.875 and 0.909. Thus, tree-based methods (RF, XGB) proved strongest, while SVM and KNN were less effective. The ROC curves for all five single classifiers are presented in Fig. 3.

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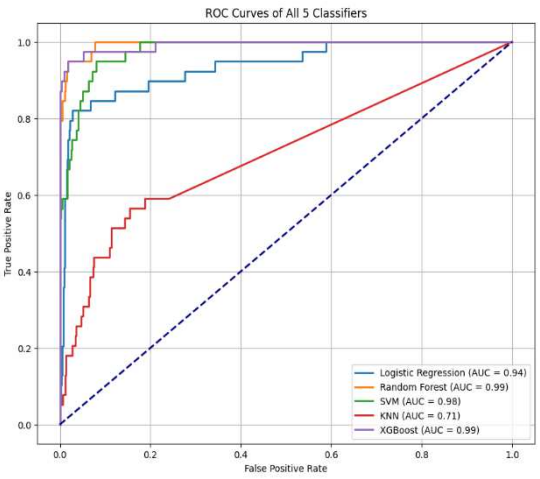


Fig. 3. ROC curve for all five single classifiers.

Building on these single-classifier results, ensemble experiments were conducted across fifteen cases with resampling and feature optimization. The baseline ensemble on the original dataset achieved 0.975 accuracy with precision, recall, and F1 near 0.974–0.975 (Table III). Random oversampling without optimization improved performance to 0.988, confirming that expanding the minority class enhances detection. By contrast, random undersampling reduced accuracy to 0.923 due to loss of majority samples.

Oversampling with decision tree–based feature importance (Case 6) produced the best results (0.999 across all metrics). XGB feature importance (Case 7) achieved 0.988, while PCA (Case 8) and LDA (Case 9) yielded 0.980 and 0.961. Feature selection methods (mRMR, RFE) under oversampling were least effective, with accuracies near 0.761.

With undersampling, the best outcome came from RFE (Case 11), reaching 0.920, followed by DT and XGB feature importance (Cases 12 and 13) at 0.942 and 0.943. The accuracy from feature reduction was moderate. PCA achieved 0.875, while LDA reached 0.903.

TABLE I. PERFORMANCE OF SINGLE CLASSIFIERS (10-FOLD CROSS-VALIDATION)

Classifier	Accuracy	F1-score	Precision	Recall
LR	0.96	0.95	0.96	0.95
RF	0.98	0.98	0.98	0.98
SVM	0.936	0.875	0.936	0.906
KNN	0.934	0.909	0.934	0.914
XGB	0.98	0.98	0.98	0.98

C. Hyperparameter tuning

For hyperparameter optimization, we used RandomizedSearchCV with 3-fold cross-validation. The parameter ranges for each model are in Table II. We chose this method instead of a grid search to reduce computational costs while still effectively covering the hyperparameter space.

TABLE II. HYPERPARAMETER TUNING RESULTS

Model	Tuned Parameters (Range)	Best Parameters Found
Logistic Regression	C = {0.01, 0.1, 1, 10}, solver = {liblinear, lbfgs}	C=1, solver=liblinear
Random Forest	n_estimators = {50,100,200}, max_depth={None,5,10}	n_estimators=200, max_depth=10
SVM	C={0.1,1,10}, kernel={linear,rbf}, gamma={scale,auto}	C=10, kernel=rbf, gamma=scale
KNN	n_neighbors={3,5,7,9}, weights={uniform,distance}	n_neighbors=7, weights=distance
XGBoost	n_estimators={50,100,200}, max_depth={3,5,7}, learning_rate={0.01,0.1,0.3}	n_estimators=100, max_depth=5, learning_rate=0.1

TABLE III. PERFORMANCE OF EXPERIMENTAL CASES USING ENSEMBLE ML MODEL (10-FOLD CROSS-VALIDATION)

Case No.	Sampling	Feature Optimiz-ation	Accu-racy	Precision	Recall	F1-Score
1	Original	–	0.975	0.974	0.975	0.974
2	RO	–	0.988	0.997	0.997	0.997
3	RUS	–	0.923	0.928	0.923	0.923
4	RO	Feature Selection (mRMR)	0.761	0.763	0.762	0.762
5	RO	Feature Selection (RFE)	0.762	0.763	0.762	0.762
6	RO	Feature Importance (DT)	0.999	0.999	0.999	0.999
7	RO	Feature Importance (XGB)	0.988	0.989	0.988	0.988
8	RO	Feature Reduction (PCA)	0.980	0.981	0.980	0.980
9	RO	Feature Reduction (LDA)	0.961	0.961	0.961	0.961
10	RUS	Feature Selection (mRMR)	0.682	0.689	0.682	0.681
11	RUS	Feature Selection (RFE)	0.920	0.924	0.920	0.920
12	RUS	Feature Importance (DT)	0.942	0.946	0.942	0.942
13	RUS	Feature Importance (XGB)	0.943	0.945	0.943	0.943
14	RUS	Feature Reduction (PCA)	0.875	0.879	0.875	0.875
15	RUS	Feature Reduction (LDA)	0.903	0.907	0.903	0.903

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D. Soft versus Hard Voting

For the top-performing cases, both soft voting and hard voting ensemble strategies were evaluated in Table IV. Across all tested configurations, soft voting consistently outperformed hard voting. For example, in Case 6, soft voting achieved 0.999 accuracy compared to 0.9819 for hard voting. Similar trends were observed for Cases 7 and 8, where soft voting improved accuracy by approximately one percentage point over hard voting. This demonstrates that probability averaging in soft voting better leverages the strengths of heterogeneous classifiers, particularly in imbalanced classification scenarios. In Fig. 4, the ROC curve for the top hard voting model is shown.

TABLE IV. HARD VOTING RESUL FOR TOP-PERFORMING CASES

Top Cases	Voting Type	Accuracy	Precision	Recall	F1-Score
RO- Feature Importance (DT)	Hard	0.9819	0.9825	0.9819	0.9819
RO- Feature Importance (XGB)	Hard	0.9801	0.9809	0.9801	0.9801
RO- Feature Reduction (PCA)	Hard	0.9686	0.9705	0.9686	0.9686

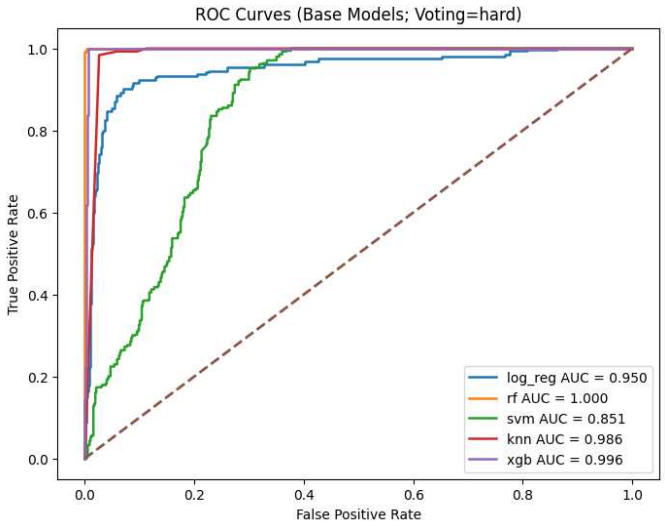


Fig. 4. ROC CURVE (Hard Voting) for the best performed model.

E. Interpretation and Clinical Relevance

The results confirm that integrating oversampling with tree-based feature importance and a soft-voting ensemble yields superior predictive performance for thyroid disease classification. The near-perfect recall in the best-performing configuration is particularly important in clinical contexts, where missing a sick case can have severe consequences. Additionally, explainable AI was used to improve the reliability of the proposed model. As shown in Fig. 5, the SHAP (SHapley Additive exPlanations) analysis shows the role of individual clinical features and their effect on the model's output. The application of explainable artificial intelligence to the best model provides interpretability, enabling clinicians to understand the contribution of individual features to predictions, thereby enhancing trust and facilitating adoption in decision-support systems.

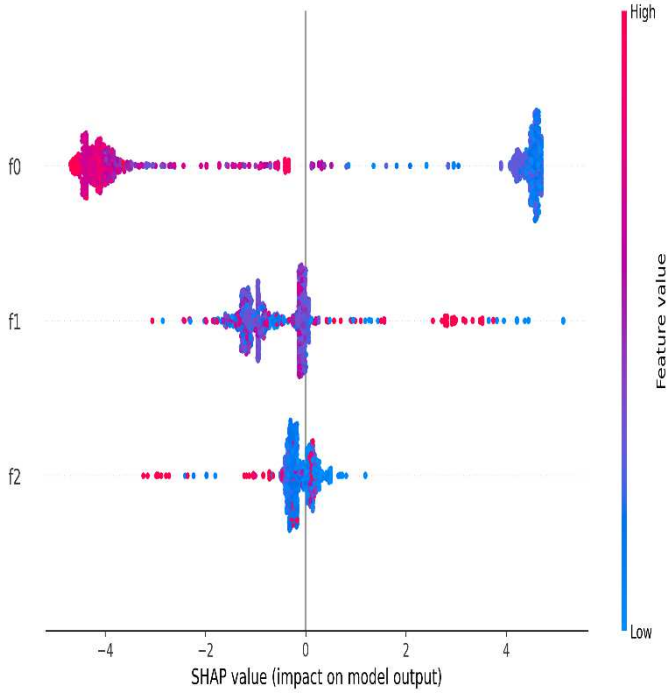


Fig. 5. Explainable AI (SHAP) Analysis for the Best Performed Model

V. DISCUSSION

The proposed model demonstrated that RO combined with Decision Tree–based feature importance and a soft-voting ensemble achieved the highest performance, with 0.999 accuracy, precision, recall, and F1-score under 10-fold cross-validation. It surpassed Uddin et al. and Raza et al., who employed ensemble learning with oversampling. Unlike Zhang et al.'s CNN, which needed special tuning for imaging, the tabular approach worked well with a variety of clinical data. Unlike Zhang et al.'s CNN that required imaging-specific tuning, the tabular approach generalized well across clinical data. Traditional classifiers from Chaganti et al. lacked robustness under imbalance, while our results confirm minority class detection improves with oversampling and tree-based optimization. Explainable AI further enhances interpretability. Overall, the pipeline surpasses prior models while balancing predictive accuracy and clinical relevance, making it effective for thyroid disease diagnosis. A comparative summary with related studies is provided in Table V.

TABLE V. COMPARISON RESULTS OF OUR MODEL WITH OTHER STUDIES.

Reference Study	Dataset	Model	Accuracy	XAI
Uddin et al. [7]	Thyroid Disease	Ensemble Model	99.71%	No
Raza et al. [8]	Thyroid Sickness Determination	LGBM	96%	Yes

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Zhang et al. [9]	Thyroid imaging dataset	CNN	97.2%	No
Chaganti et al. [10]	Thyroid Disease	SVM, KNN, DT, NB, RF	90%	No
Proposed Work	Thyroid Sickness Determination [14]	Ensemble Model	99.9%	Yes

VI. Conclusion

This study proposed an ensemble-based machine learning framework that integrates sampling strategies with feature optimization for classifying thyroid disease into negative and sick categories. A dataset with class imbalance was used to evaluate 15 cases involving Random Oversampling (RO), Random Undersampling (RUS), and three feature optimization strategies: selection, importance, and reduction. Combining RO with Decision Tree-based feature importance and a soft-voting ensemble produced the best results, with accuracy, precision, recall, and F1-score all hitting 0.999 under 10-fold cross-validation. When it came to aggregating probabilistic outcomes, soft voting consistently outperformed hard voting. For early thyroid identification, the framework offers a clinically useful and comprehensible decision-support tool. Future work will validate the model on external datasets, investigate cost-sensitive learning, and enhance explainability with finer XAI analyses.

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